Stark Shifts in Simulated Excited State Vibrational Spectra

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Abstract

We have modeled the effect of external electric fields on nitrile vibrational signatures of *2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3- benzothiadiazol-4-yl)methylene]propanedinitrile* (DTDCPB). DTDCPB is a promising push-pull chromophore prevalent in organic photovoltaic research. By applying static electric fields *in silico* and simulating ground state IR spectra, we determined the Stark tuning rate of DTDCPB nitrile stretching to be FILLINTHEBLANK. Born-Oppenheimer molecular dynamics was used to derive the model IR spectra to correlate to experimental data. In hopes of modeling pump-probe transient IR studies of DTDCPB, computational studies of excited-state dynamics are underway. These studies will enable *in operando* monitoring of electric fields of organic photovoltaic devices.

Graphical TOC Entry

Introduction

As research in solar energy becomes increasingly important, several new strategies are being investigated to increase the efficiency of these devices. In organic photovoltaics (OPVs), one new approach uses small molecules with large ground-state dipole moments as electron donors. It is believed the inherent dipole allows for enhanced hole hopping and can minimize charge recombination (ref Griffith). One such molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene] propanedinitrile (DTDCPB) and has resulted in OPVs with power conversion effiencies of up to 9.6%. The nitrile groups present in the molecule can act as reporter moieties for vibrational spectroscopy. By modeling the effect of an applied electric field on absorption spectrum (Stark Effect), it is possible to probe the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. This first requires a thorough understanding of the electric field dependence of the groundand excited state IR spectra.

Understanding the mechanisms of charge transfer events enables rational design and enhancement of current materials for novel, high-efficiency semiconductor and photovoltaic devices. First, frequency calculations determined the equilibrium properties of DTDCPB in *Gaussian 09*. The Stark effect was modeled by frequency calculations in external static electric fields of various magnitudes. Spectral shifts were obtained from these frequency calculations and were correlated to the presence of an electric field. Subsequently, Born-Oppenheimer Molecular Dynamics were used to obtain model spectra using a different set of methods from the standard frequency calculations in *Gaussian 09*.

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Theory

Standard Frequency Calculations

The *normal modes of vibration* are calculated by diagonalizing a matrix of the second spatial derivatives of the electronic potential energy surface (PES) for the system of N atoms. Diagonalization of the Hessian matrix yields the orthogonal set of N frequency normal modes $\{\omega_k\}$. The eigenvalues of this Hessian correspond to the normal modes of vibration for the system. Additionally, the anharmonicity tensor A is defined by the third spatial derivative of the PES and may be used for simulating Raman spectra. To obtain the intensities of the transitions the vibrational intensities are given by (Equation 1). The dipole moment μ is obtained from the instantaneous electron density around the nuclei.

$$\mathcal{A}_{\tilde{\nu}}(q_k) \propto \left(\frac{\partial \mu}{\partial q_k^2}\right) \tag{1}$$

Scaling the frequencies $\{\omega_k\}$ by the intensity $A_{\tilde{\nu}}(q_k)$ a simulated vibrational spectrum for the molecule.

The Stark Effect

The Stark effect is a phenomenon in which the energy level splittings are increased in the presence of an applied electric field. The effect is measured via changes to the vibrational and optical spectra of a species **(Figure?)**. We can also model it computationally by performing calculations accounting for the presence of an external field. The observables are a change in dipole moment ($\Delta\mu$) dependent on the degree of charge separation for a given transition and a change in polarizability ($\Delta\alpha$) corresponding to the change in sensitivity of the vibrational transition to the applied field. The change in frequencies is modeled in Equation 2 where $V(r_k)$ is the potential with respect to position r_k of electron k.

$$\Delta\omega_k = -\Delta\vec{\mu} \cdot \left(\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\right) - \frac{1}{2}\Delta\alpha \cdot |\frac{\partial V(\vec{r})}{\partial \vec{r}_k}\Big|^2 \tag{2}$$

Born-Oppenheimer MD (BOMD)

We apply Born-Oppenheimer Molecular Dynamics (BOMD) to derive the model spectra for DTDCPB with and without the static electric field present. BOMD utilizes Born-Oppenheimer approximate time-*independent* density functional theory (DFT) calculations to compute the electronic potential at each nuclear evolution time step with classical equations of motion (EOMs) integrated by the Velocity Verlet integration method (Equations 3) approximating the nuclear dynamics.

$$\begin{aligned} \vec{x}(t+\Delta t) &= \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2 \\ \vec{v}(t+\Delta t) &= \vec{v}(t) + \frac{\vec{a}(t)+\vec{a}(t+\Delta t)}{2}\Delta t. \end{aligned} \tag{3}$$

In this context, BOMD has advantages over the *ab initio* frequency calculation discussed above. The Stark effect is most readily observed in vibrational spectra, so using purely *ab initio* techniques we require either the second or third derivative of the electronic PES, which can be computationally expensive for larger systems. In contrast, BOMD only requires the gradient of the electronic PES to propagate nuclear position (Equations 3) and only the electron density to compute the time-dependent dipole moment (Equation 4). Thus, BOMD manages to capture some anharmonicity and higher order terms of the potential $V(\vec{r}_k)$, (Equations **??** and **??**) with only the gradient for each time step.

$$\vec{\mu}(t) = -\sum_{mu}\sum_{nu} P_{\mu\nu}\big(\nu|\mathbf{r}(t)|\mu\big) + \sum_{A} Z_A \mathbf{R}_A(t) \tag{4}$$

Obtaining Spectra from Dipole Orientation Time Series

Since BOMD only calculates the first derivative of the PES, another method is needed to obtain spectra. The dipole vector $\vec{\mu}(t)$ is output from BOMD as a time series, so taking the first time derivative $\dot{\vec{\mu}}(t)$ and using the *dipole autocorrelation function* $\langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau$ as the kernel of an integral transform (Equation 5 we obtain the normal mode frequencies and intensities.

$$\mathcal{A}(\omega) \propto \int \langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t+\tau)\rangle_\tau e^{-i\omega t} dt. \tag{5}$$

Plotting the function $\mathcal{A}(\omega)$ along the frequency domain yields a model spectrum. (**518 words**)

Methodology

All *ab initio* calculations performed in this work used the B3LYP/6-311G(d,p) level of theory. The static field was chosen rather than an oscillating field to simulate the local environment around each DTDCPB dimer immediately after another nearby dimer was stimulated into a charge transfer state. First, the standard frequency calculations for the ground state of DTDCPB both with and without an applied static field were performed to determine the equilibrium geometries in the various environments and generate IR spectra. The Stark effect was modeled by applying a static dipole oriented along the z-axis normal to the plane in which the majority of the DTDCPB molecule was embedded. The spectra for the equilibrium geometries of the ground state in various field strengths were compared and tabulated to verify that the Stark effect was captured by the model and could be "calibrated" for field strength.

BOMD was first applied to a cyanide ion as a test case to coarsely model the nitriles on DTDCPB. Subsequently, it was applied to DTDCPB monomer in zero field and also in an applied field. The systems were brought to ideal initial conditions for the "Production MD" steps by performing thermal Monte Carlo sampling subject to Boltzmann statistics in the same electric field to generate multiple trajectories for nuclear dynamics with thermally populated vibrational normal modes. Finally, the production MD runs yielded the dipole time series for calculating the spectra. The individual spectra from the multiple production MD runs can be averaged in the frequency domain to obtain a single spectrum from multiple BOMD runs. Our BOMD application scheme follows that outlined in Figure **??** (**274 words**)

Computational Results

Discussion

Conclusion

Based on the computational results, there appears to be a clear Stark shift occurring in the ground state spectra which is nearly symmetric with respect to inversion of applied field orientation. The low-intensity signals corresponding to the asymmetric nitrile stretching frequencies echo the Stark shift occurring in the stronger symmetrical stretching signal as well. Furthermore, we found that BOMD could be applied to the cyanide and maintain proper energy conservation in both the zero field and applied field cases. This implies that we are now able to extend this work to study the entire DTDCPB dimer system with and without applied fields in both ground and excited states, which forms the basis for our continuing work. Once the BOMD runs are successfully completed, the spectra can be compared with experiment to determine the accuracy of this technique for this type of system. **142 words**

Acknowledgement

We would like to thank and acknowledge Professor Xiaosong Li and Professor Cody Schlenker for their support and invaluable feedback. We would also like to thank the members of the Li Research Group for making themselves available to answer our computational questions and helping us troubleshoot *Gaussian*. Special thanks go to Dr. Alessio Petrone, Dr. Bo Peng, David Lingerfelt, Patrick Lestrange, and Hongbin Liu for spending their valuable time helping us with this work. Additionally, we would like to thank Hyak, The University of Washington IT Services, and the High Performance Computing Club for making available to us the computational resources required to perform this work.